

EXPERIMENT 3: INPUT & OUTPUT CHARACTERISTICS OF AN N-CHANNEL MOSFET (2N7000)

1. Aim

- To plot the output (drain) characteristics of an NMOSFET (I_D vs V_{DS} for several fixed values of V_{GS}), and to determine the drain (output) resistance r_d in the saturation region.
- To plot the transfer (input) characteristics of an NMOSFET (I_D vs V_{GS} at fixed V_{DS}), and to determine the threshold voltage V_T and the transconductance g_m .
- To compute the amplification factor $\mu = g_m \times r_d$ from the measured data.

2. Apparatus / Components Required

S.No.	Component / Equipment	Specification	Quantity
1	2N7000 transistor	N-channel enhancement MOSFET, TO-92	1
2	Regulated DC power supply	0–30 V, dual output, adjustable current limit	2
3	Milliammeter	0–50 mA range	1
4	Voltmeter	0–30 V range	2
5	Fixed resistors	1 k Ω , 4.7 k Ω , 10 k Ω	as needed
6	Breadboard, connecting wires, probes	—	as needed

3. Device Ratings and Safety Limits

The 2N7000 is rated for a much wider voltage and current range than this experiment will use — the limits below are deliberately kept well inside the datasheet maximums so that normal variation between individual devices (see the threshold-voltage spread noted below) cannot push any device into damage.

3.1 Absolute Maximum Ratings (from datasheet)

Parameter	Symbol	Rating
Drain–source voltage	V_{DS}	60 V
Gate–source voltage (continuous)	V_{GS}	± 20 V
Drain current (continuous)	I_D	200 mA
Drain current (pulsed)	$I_D(\text{pulse})$	500 mA
Total power dissipation ($T_A = 25^\circ\text{C}$, TO-92, no heatsink)	PD	400 mW (derates 3.2 mW/ $^\circ\text{C}$ above 25°C)
Gate threshold voltage (device-to-device spread)	$V_{GS}(\text{th})$	0.8 V to 3.0 V (typ. ≈ 2.1 V)

3.2 Safe Working Limits for This Lab

Stay within the following limits at all times — they hold instantaneous power dissipation ($V_{DS} \times I_D$) to roughly half of the 400 mW rating, leaving headroom for the fact that a hand-wired TO-92 on a breadboard has no heatsink:

- V_{DS} : do not exceed 10 V.
- V_{GS} : do not exceed 5 V.
- I_D : do not exceed 30 mA — before powering the circuit, set the power supply's current-limit (compliance) control to 30 mA, so the supply itself clamps the current if a device turns on more sharply than expected.
- Because $V_{GS(th)}$ varies from 0.8 V to 3.0 V across individual devices, never assume a fixed V_{GS} will give a fixed I_D — always watch the milliammeter as you raise the voltage, and stop raising it once I_D reaches the 30 mA limit, regardless of what voltage that occurs at.
- Take each reading quickly; do not leave the device biased at high V_{DS}/I_D for longer than needed to note the value.

4. Theory

4.1 MOSFET Fundamentals

A Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET) is a three-terminal device — gate (G), drain (D), and source (S) — in which the gate is electrically insulated from the channel by a thin oxide layer, so the gate draws essentially no DC current ($I_G \approx 0$). In an N-channel enhancement-mode MOSFET, applying a gate–source voltage V_{GS} above the threshold voltage V_T attracts electrons to form a conducting channel between drain and source. With no channel formed ($V_{GS} < V_T$) the device is in cutoff. Once a channel exists, the device operates in the triode (ohmic) region for small V_{DS} , where I_D rises roughly linearly with V_{DS} , and in the saturation region once $V_{DS} \geq V_{GS} - V_T$ (pinch-off), where I_D becomes almost independent of V_{DS} and follows the square law $I_D \approx k(V_{GS} - V_T)^2$. A slight upward slope in the saturation region (finite output/drain resistance) is due to channel-length modulation in real devices.

4.2 Output (Drain) Characteristics

The output characteristic (I_D vs V_{DS} at fixed V_{GS}) shows the triode region merging into the flatter saturation region at the pinch-off point $V_{DS} = V_{GS} - V_T$. The locus of these pinch-off points across different V_{GS} traces the boundary between the two regions. The reciprocal of the slope of the (nearly flat) saturation-region curve gives the drain resistance, $r_d = \Delta V_{DS} / \Delta I_D$. The curves below are illustrative, plotted only up to this lab's 30 mA safety ceiling.

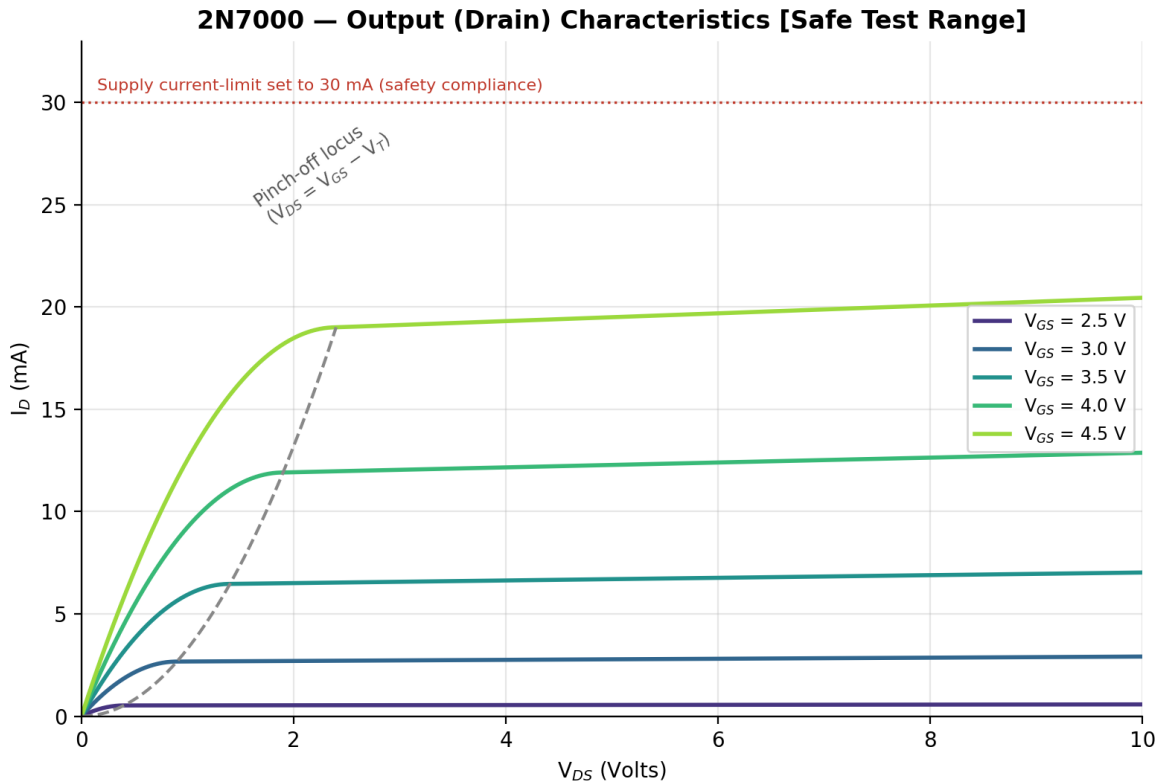


Fig. 1: Illustrative 2N7000 output (drain) characteristics within this lab's safe test range — I_D vs V_{DS} for several fixed values of V_{GS} , with the pinch-off locus shown dashed.

4.3 Transfer (Input) Characteristics

The transfer characteristic (I_D vs V_{GS} at fixed V_{DS} , with V_{DS} held high enough to keep the device in saturation throughout the sweep) shows no drain current until V_{GS} exceeds the threshold voltage V_T , after which I_D rises quadratically with $(V_{GS} - V_T)$. The slope of this curve, $g_m = \Delta I_D / \Delta V_{GS}$, is the transconductance — a key small-signal parameter of the device. Because V_T varies from device to device (0.8–3.0 V), your measured curve may be shifted left or right of the illustration below — that is expected.

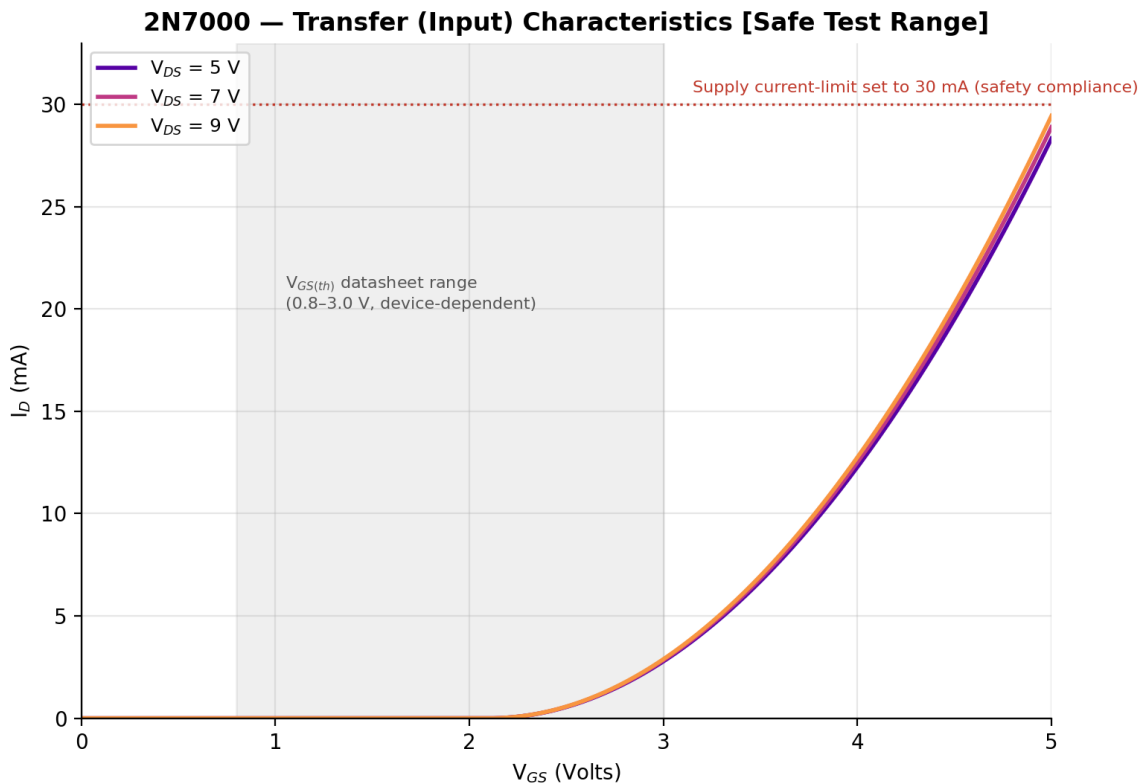


Fig. 2: Illustrative 2N7000 transfer (input) characteristics within this lab's safe test range — I_D vs V_{GS} for several fixed values of V_{DS} .

5. Part A — Output (Drain) Characteristics

Circuit Connections

1. Connect the MOSFET source to ground (common). Connect the drain through a milliammeter (0–50 mA) to the drain supply V_{DD} , and connect a voltmeter (0–30 V) across drain–source.
2. Connect the gate directly to the gate supply V_{GG} (no series resistor needed, since $I_G \approx 0$), with a voltmeter (0–30 V) across gate–source.
3. Before switching on, set the drain supply's current-limit (compliance) control to 30 mA.

Procedure

4. Set $V_{GS} = 2.5\text{ V}$ and hold it fixed for this run.
5. Starting with $V_{DS} = 0\text{ V}$, increase V_{DS} gradually — in steps of 0.2 V up to 2 V (to trace the triode region accurately), then in steps of 1 V beyond that — recording I_D at each step. Stop increasing V_{DS} once you reach 10 V, or sooner if I_D reaches the 30 mA compliance limit first.
6. Repeat step 1 for $V_{GS} = 3.0, 3.5, 4.0$, and 4.5 V , each time stopping the V_{DS} sweep at 10 V or at the 30 mA current limit, whichever comes first.
7. If, for any V_{GS} , I_D hits the 30 mA limit at a low V_{DS} , record that as the last valid point for that curve and move on — do not increase V_{GS} further to try to reach a higher V_{DS} at that current.
8. Switch off the supply. Plot I_D (Y-axis) vs V_{DS} (X-axis) as a family of curves, one for each V_{GS} .

Observation Table — Output (Drain) Characteristics

S.No.	VDS (V)	ID (mA) at VGS=2.5V	ID (mA) at VGS=3.5V	ID (mA) at VGS=4.5V
1				
2				
3				
4				
5				
6				
7				
8				
9				

6. Part B — Transfer (Input) Characteristics

Circuit Connections

- Retain the circuit of Part A, with the 30 mA current-limit still set on the drain supply.

Procedure

- Set $V_{DS} = 5\text{ V}$ and hold it fixed for this run (high enough to keep the device in saturation across the sweep).
- Increase V_{GS} gradually from 0 V, watching the milliammeter continuously. Expect $I_D \approx 0$ until V_{GS} reaches somewhere in the 0.8–3.0 V range (this varies by device) — do not assume a fixed turn-on voltage.
- Once I_D starts rising, slow down to 0.1 V steps and continue recording I_D vs V_{GS} until either V_{GS} reaches 5 V or I_D reaches the 30 mA compliance limit, whichever comes first. Do not exceed $V_{GS} = 5\text{ V}$ under any circumstances.
- Repeat steps 2–3 for $V_{DS} = 7\text{ V}$ and $V_{DS} = 9\text{ V}$.
- Switch off the supply. Plot I_D (Y-axis) vs V_{GS} (X-axis) as a family of curves, one for each V_{DS} .

Observation Table — Transfer (Input) Characteristics

S.No.	VGS (V)	ID (mA) at VDS=5V	ID (mA) at VDS=7V	ID (mA) at VDS=9V
1				
2				
3				
4				
5				
6				
7				
8				

9				
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7. Calculations / Analysis

- From the transfer characteristic: identify the threshold voltage V_T (the V_{GS} at which I_D first becomes measurable for your specific device), and compute the transconductance $g_m = \Delta I_D / \Delta V_{GS}$ at a chosen operating point in the saturation region.
- From the output characteristic: compute the drain (output) resistance $r_d = \Delta V_{DS} / \Delta I_D$ from the slope of the saturation-region curve at fixed V_{GS} .
- Compute the amplification factor $\mu = g_m \times r_d$.
- Compare the measured pinch-off points (V_{DS} at which each output curve flattens) against $V_{GS} - V_T$ computed from the transfer characteristic's V_T .
- Optionally, plot $\sqrt{I_D}$ vs V_{GS} from the transfer data — it should be linear in saturation, and its X-intercept gives an independent estimate of V_T .